



A novel financial performance metric to minimize misclassification costs in model selection

John Martin¹ · Mali Abdollahian¹ · Sona Taheri¹ · David Akman¹

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Abstract

A novel financial performance metric (FPM) is introduced seeking to minimise the misclassification cost arising from false positives and false negatives in credit risk assessment. Using the German Credit Dataset (GCD), important financial variables are simulated according to four different asset classes to enable a more accurate and reliable, multidimensional model selection. The misclassification cost arising from FPM is compared with commonly used statistical metrics and the credit scoring example dependent cost matrix (CSEDCM) metric. The results show that CSEDCM underestimates false prediction costs by as much as 99% compared to the FPM. A range of high- performance machine learning methods was compared using FPM and statistical metrics. The Multi-Layer Perceptron outperformed other methods on statistical metrics and overall on financial costs, while a mix of algorithms worked best on either side of the decision threshold. The results confirmed that the proposed FPM would provide a significant financial benefit to organisations.

Keywords Credit scorecard · Financial performance metric · Machine learning · German credit

1 Introduction

A credit default is a negative event for both borrowers and lenders, but can also impact the broader economy. Because credit is a key driver of the economy, essential for the stability and growth of the global financial system, no economy, irrespective of how advanced it may be, can develop in its absence (Banu, 2013). As the global financial crisis and more recent events such as the COVID-19 pandemic have demonstrated, credit losses related to loans can have a significant detrimental impact on the broader economy (Cantrell et al., 2014).

✉ Sona Taheri
sona.taheri@rmit.edu.au

John Martin
S3801949@student.rmit.edu.au

Mali Abdollahian
mali.abdollahian@rmit.edu.au

David Akman
david.akman@rmit.edu.au

¹ School of Science, RMIT University, Melbourne, VIC, Australia

To protect against losses, lenders seek to maintain robust lending practices through optimising credit risk assessment. Indeed, the centrality of credit to the monetary economy throughout the economic cycle underscores the importance of accurate credit assessment. Historically, creditworthiness was evaluated using human judgement. However, the scale of lending has increased over time such that contemporaneously, most lenders deploy classification models to perform this task at scale.

Scorecard models are one of the main applications of classification models in credit risk. Used to assess creditworthiness, scorecard models associate historical characteristics like prior loan performance with default to estimate the probability of default. Once this has been derived it is then compared to cut-off thresholds to determine the outcome: approval or rejection.

A great deal of prior research has been published detailing the comparative performance of various statistical and machine learning algorithms. Martin et al. (2024) compared a wide range of individual and ensemble algorithms for credit scoring. They showed that the best individual performer was a Generalised Additive Model and the best overall performer was a Random Forest and K-Nearest Neighbour ensemble. Similarly, hybrid models have performed well on public datasets such as German credit. Tripathi et al. (2020) used an Extreme machine learning approach. Mahbobi et al. (2023) found that the Support Vector Machine algorithm performed best under a K-Nearest Neighbour sampling approach.

We consider methods which have recently been shown to outperform other approaches in credit risk modelling research including Random Forest (RF), Gradient Boosting Machine (GBM), Artificial Neural Networks, specifically the Multi-Layer Perceptron (MLP), Decision Trees (DTR) and Support Vector Machine (SVM). We compare the performance of these five algorithms with that of the gold standard used in industry, Logistic Regression (LR), using the statistical and cost performance criteria discussed in the next sections.

Main contributions of this paper are:

1. Introducing a new Financial Performance Metric (FPM) to estimate the true costs arising from both false positives and false negatives;
2. Providing insights on cost dynamics as a consideration in the model selection, and optimising the selection by combining statistical methods with the developed FPM;
3. Evaluating the suitability of a publicly available German Credit Dataset (GCD) for credit risk modelling, and simulating important financial variables which are often not available in public datasets;
4. Comparing the proposed FPM with an existing metric, credit scoring example dependent cost matrix (CSEDCM) metric, as well as commonly used statistical criteria;
5. Comparing cost dynamics across four simulated asset classes observed at large banks to understand the impact of default and opportunity cost across different loan values;
6. Comparing the performance of Logistic Regression with a selection of high-performance machine learning algorithms utilizing the commonly used statistical criteria and the proposed FPM, and determining which algorithm outperforms in estimating costs arising from both false positives and false negatives.

The rest of the paper is organised as follows. First, we provide a brief introduction on the research work in Sect. 1. In Sect. 2, we present an overview of commonly used statistical measures of performance. Section 3 presents an overview on cost performance measures and introduces a new approach. In Sect. 4, the machine learning methods used are provided. The dataset and numerical results are described in Sect. 5. The discussion, conclusion and some future direction are provided in Sect. 6.

2 Statistical performance

Traditionally the focus in credit risk modelling research has been to optimise the identification of true positives, that is, optimizing alignment between the prediction and observation of default, with scant regard for ‘non-defaults’. Researchers and industry practitioners alike have mainly relied on the use of statistical metrics to measure model performance and select the strongest model.

Conceptually, the underlying logic to this practice is that by selecting the model with the strongest metrics modellers can identify the model which is most fit for purpose. In industry and academia alike, classical statistical performance criteria including the Area under the Receiver Operating Curve (AUC), Accuracy (ACC) and Somers’ Delta (Gini) are amongst the most widely used model metrics reported and used for model selection.

However, prior research has identified some problematic features of AUC, ACC and Gini, which suggest their use can lead to incorrect conclusions (see, for example, Verbraken et al., 2014). The Gini statistic has a linear relationship with AUC, yet both are commonly reported despite the obvious redundancy of reporting directly related performance metrics. Gini is defined as

$$Gini = 1 - \sum_{k=2}^{n+m} \left[(F_{m.BAD_k} - F_{m.BAD_{k-1}}) \times (F_{n.GOOD_k} - F_{n.GOOD_{k-1}}) \right],$$

where $F_{m.BAD_k}$ ($F_{n.GOOD_k}$) is the k^{th} vector value of the empirical distribution function of bad (good) applicants, while m and n are the number of elements of the bad and good distributions, respectively. Further details can be found, for example in Thomas et al. (2002).

Hand (2009) states that Gini may not be suitable as a measure of performance of application scorecards since it (as well as some other commonly used metrics including the Kolmogorov–Smirnov (KS) statistic and Information Value (IV)) uses irrelevant information. It fails to measure information on the bad rate among accepts which is the aspect of performance we are actually interested in.

The Receiver Operating Curve (ROC) is a graphical tool used to visualise the trade-off between sensitivity and specificity, while the AUC is a scalar measure of aggregate performance which summarises the ROC into a single value. In Anagnostopoulos et al. (2019), the AUC is criticised for how it handles the trade-off between false positives and false negatives whereby the relative severities of misclassifications are treated differently between different classifiers. Moreover, Marzban (2004) showed that AUC discriminates well between ‘good’ and ‘bad’ models, but not between good models. Hand and Anagnostopoulos (2013) dismiss AUC as a portmanteau measure, equivalent to integrating over a range of possible values, concluding that the ROC is an incoherent performance measure. The AUC is defined as

$$AUC = \frac{1}{2} \times (Sensitivity + Specificity),$$

where sensitivity is the True Positive Rate (TPR) and specificity is the True Negative Rate (TNR), with Sensitivity (TPR) = True Positives/Total Positives $\times 100$, whereas Specificity (TNR) = True Negatives/Total Negatives $\times 100$.

As stated above, there is a linear relationship between AUC and Gini, and each can be easily calculated from the other using the following equation, suggesting redundancy in the

common practice of presenting both measures for model evaluation and/or selection:

$$AUC = \frac{Gini + 1}{2}.$$

On the other hand, ACC works by effectively summing accurate classifications as a proportion of all classifications, which is problematic in scenarios characterised by severely imbalanced target variables typical in credit modelling research. For example, for an observed default rate of 0.05 a model may predict *all* observations as non-defaults, yet still receive an ACC of 0.95. While a common solution is to perform sampling to increase the minority class representation. This rarely fully deals with the class imbalance bias in real-situations and creates further complications downstream in terms of model monitoring. The ACC is defined as

$$ACC = \frac{TP + TN}{TP + TN + FP + FN},$$

where TP is equal to True Positive, TN equals True Negative, FP equals False Positive and FN equals False Negative.

3 Cost performance

As discussed in the previous section, the most commonly used statistical metrics have some serious drawbacks which raises questions around their value in model selection, particularly in relation to commercial applications such as credit risk. Moreover, focusing only on this aspect of performance ignores some very important practical considerations. In classification research the correct selection of performance metrics is one of the most important issues in evaluating a classifier's performance (Liu et al., 2014). Indeed, a vast array of alternative metrics are available and optimizing the wrong metric directly translates into lost revenue (Dmitriev & Wu, 2016).

The most important objective in banking is profit, and therefore once a model has been determined as functional the comparative costs must be considered. Selecting the optimal approach is not possible using standard classification metrics because they treat the costs of misclassifications the same, which is not true in real credit risk management (see, for example, Fiore et al., 2017). Credit risk is particularly well aligned to the use of profit-based or financial loss-related measures (Maldonado et al., 2017), which seek to optimise a commercial outcome as opposed to a statistical metric.

Cost sensitive learning seeks to optimise decision-making where misclassification costs incur different penalties. While a valid application of cost sensitive learning can assume the same misclassification cost, Elkan (2001) suggested that a more realistic problem exists where misclassification costs are example-dependent, in the sense that the costs vary among examples and not only among classes. This is particularly so for credit risk, where misclassification costs vary both within and between response classes.

Indeed, researchers have long been aware of the profit motive as a central driving factor in model selection in credit risk yet only a handful have sought to find ways to incorporate this perspective into model selection. One of the biggest obstacles has been the dearth of available datasets containing the necessary financial information to compare performance on cost (Aodha & Brostow, 2013), leading to a variety of approaches to overcome this shortfall. For example, Xia et al. (2017) simply assumed that the costs of misclassifying a defaulting borrower are larger than that of misclassifying a good one. Schebesch and Stecking (2005)

Table 1 CSEDCM model, given in Eq. (1)

	Actual positive $y_i = 1$	Actual negative $y_i = 0$
Predicted positive $c_i = 1$	$C_{TP_i} = 0$	$C_{FP_i} = r_i + C_{FP}^a$
Predicted negative $c_i = 0$	$C_{FN_i} = Cl_i \cdot LGD$	$C_{TN_i} = 0$

simplified this problem by assuming that the misclassification costs of a bad borrower as a good borrower as five times more costly than misclassification of a good borrower as a bad borrower, yet without data neither paper was able to empirically evaluate this assumption.

Other researchers with access to data containing financial performance variables have constructed cost metrics using a variety of different approaches. For instance, Zhang et al. (2018) applied the Multiple Instance Learning method, proposing a cost sensitive optimization approach which sought to minimise misclassification costs. In this work, instead of quantifying differences in misclassification costs, the authors weighted misclassifications by similarity according to demographic features and transactional behaviour.

3.1 Existing cost metrics

Wang et al. (2021) proposed an approach to estimate costs associated with risk based pricing, whereby similar groups of exposures were assigned increasing levels of interest relative to their risk in peer-to-peer lending. They assigned a cost matrix to pooled risks for differential pricing, which sought to assign differential misclassification cost matrices, with the lower triangular C1 assigned the economic costs with a predicted ‘good defaulting’, and C2 the additive cost of misclassifying an applicant to the wrong pool and the opportunity cost of lost business. Nevertheless, the uncertainty arising from new business in opportunity costs was not considered in this work.

By contrast, Bahnsen et al. (2015) and Verbraken et al. (2014) both sought to assign a cost matrix with differential costs according to each outcome. Bahsen et al. (2015) focused on misclassification only, where true predictions were assumed to cost nothing. Let $\{(x_i, y_i) \in R^n \times R, i = 1, \dots, N\}$ be a set of predictor (explanatory) variables x and response (indicator) variable y . The Bahnsen et al. (2014, 2015) credit scoring example dependent cost matrix (CSEDCM) metric is given below:

$$\begin{aligned} Cost(f(x_i^*)) &= y_i \left(c_i C_{TP} + (1 - c_i) C_{FN} \right) \\ &\quad + (1 - y_i) \left(c_i C_{FP} + (1 - c_i) C_{TN} \right), \end{aligned} \quad (1)$$

where $c_i, i = 1, \dots, N$ are predicted labels, and values $C_{TP}, C_{TN}, C_{FN}, C_{FP}$ indicate the costs of true positive, true negative, false negative, and false positive, respectively. Note that $C_{TP} = C_{TN} = 0$ as the true prediction do not accrue any misclassification cost. C_{FN} shows losses if the customer i defaults to be proportional to his credit line, whereas C_{FP} is minus profit plus the expected loss if the customer does not default.

Table 1 sets out the CSEDCM approach according to the four outcomes for a binary classification model.

Here, r_i is the loss in profit by rejecting what would have been a good customer, yet this value must be present valued. The term C_{FP}^a is minus the profit of an average alternative customer ($\bar{r}$) times the probability they will not default (π_0), plus the average credit line ($\bar{C}l$) times Loss Given Default (LGD) times the probability they will default (π_1), that is $C_{FP}^a = -\bar{r} \cdot \pi_0 + \bar{C}l \cdot LGD \cdot \pi_1$.

Taking a slightly different route, Verbraken et al. (2014) sought to estimate costs compared to a model where all applicants are approved, that is where no credit scoring takes place. More specifically, they proposed the Expected Maximum Profit (EMP) model assigning a benefit in correctly identifying a defaulter, b_0 , which is the fraction of the loan amount which is lost after default as

$$b_0 = \frac{LGD \times EAD}{A},$$

where EAD is Exposure at Default, and A is Principal Exposure. By contrast, the costs associated with misclassifying a good applicant as a defaulter are equal to the Return on Investment (ROI) of the loan, defined as follows:

$$ROI = \frac{rM}{1 - (1+r)^{-M}} - 1,$$

where r is the Interest Rate, and M is Maturity.

Further, Verbraken et al. (2014) proposed that LGD follows a uniform distribution, and the EMP metric is given below:

$$EMP = \int_0^1 P(T(\theta); b_0, ROI) \times \pi_0 F_0(t) - h(b_0) db_0,$$

where t is the defined cutoff threshold value, T is the optimal cutoff value under the given circumstances, θ is the cost-benefit ratio, $\pi_0 F_0(t)$ is a true positive case, $h(b_0)db_0$ is the probability distribution associated with correctly identifying a defaulter, and

$$P(t; b_0, ROI) = b_0 \times \pi_0 F_0(t) - ROI \times \pi_1 F_1(t),$$

with $\pi_1 F_1(t)$ indicating a false positive case.

3.2 Proposed financial performance metric

In this section, we introduce a more realistic estimate of the misclassification costs arising from credit risk, the financial performance metric (FPM). We formulate the FPM based on the prior research, in particular the contributions of Bahnsen et al. (2015) and Verbraken et al. (2014). The FPM seeks to incorporate elements which have hitherto been omitted. We summarise our approach below and highlight differences with the existing approaches.

Loss estimation in credit risk has matured significantly thanks to advances required by regulatory regimes such as IFRS9, published in 2014 (IASB, 2014). IFRS9 was the international accounting standard board's response to the financial crisis, aimed at improving the accounting and reporting of financial assets and liabilities (Gea-Carrasco, 2015). We note that while lost interest is sometimes recognised in some formulations of realised LGD, we have separated out this component to make it more explicit.

While we agree that LGD must be incorporated into cost estimates to obtain a true view of the potential loss, Bahnsen's CSEDCEM metric uses maximum credit line in all estimates of losses, yet it is clear that EAD must be evaluated since EAD is the actual amount the lender has at risk. Moreover, the CSEDCEM approach ignores the interest lost from a default which

is an important consideration, noting the profit component as a key motivation for being in the business of lending.

Furthermore, in estimating the cost of a false positive, there is some debate on how to incorporate the portfolio level uncertainty for the alternative customer (the cost of the ‘next customer’ who the model deems the credit worthy). We have chosen to incorporate an amended version of the average expected loss for the portfolio using the expected loss formula, as these costs can reasonably be expected over time and have incorporated a measure of the uncertainty of future profits. In the results section we describe a comparison between the proposed approach and the CSEDCM approach to demonstrate the model selection implications.

Similarly, Verbraken’s model presents some opportunity for refinement to improve the estimation of expected costs. For example, ROI cannot possibly be as simple as multiplying the interest rate against the principle for the loan term. The fact that there is a default rate means that a proportion of the portfolio will never reach its contractually agreed maturity term. Thus remaining months on book must be considered to accurately assess the true value lost. While averages are sometimes unavoidable where specific information isn’t available, the gross value of interest revenue lost from a mortgage customer who defaults at two months is vastly different from one who defaults at the 320th month. Moreover, the financial profiles of the underlying customers who form these two disparate scenarios will be very different. Therefore, ignoring the remaining months on book would systematically underpredict costs for identifiable customer groups.

In addition, since credit decisions that result in default and opportunity costs (that is, positive and negative misclassifications) occur simultaneously, they should be summed rather than subtracted. Finally, it is unrealistic to assume a starting position of the no credit risk model as even prior to the wide-scale adoption of classification models expert judgment was used to score applicants.

Our proposed FPM advances cost-sensitive learning as it applies to credit scorecards by developing a more realistic evaluation of the financial costs arising from each misclassification. These differences can be potentially huge when considered at a portfolio level that might comprise hundreds of thousands of customers. Splitting costs on either side of the decision boundary also imparts positive benefits. It provides the capability to optimise the threshold by minimising cost. Any systematic weakness in a chosen model could be identified and indeed overcome by potentially adding variables or even ‘beefing up’ the predictive capability by combining models. Thus, the cost of a false negative, misclassifying a bad customer as good, is equal to

$$FNC = \sum_{i=1}^N P(\hat{y}_i < t \mid y_i = 1) \times LGD \times EAD + PV(ROI \times RMOB),$$

where y_i is the observed outcome (actual output), $\hat{y}_i$ is the estimated outcome (modeled output), and $P(\hat{y}_i < t \mid y_i = 1)$ are the false negative cases, the function PV is a present value which depends on ROI and $RMOB$, with $RMOB$ being the remaining months on book, a measure of the number of months remaining on the repayment terms of the loan agreement.

On the other hand the cost of a false positive, misclassifying a good customer as bad, is equal to

$$FPC = \sum_{i=1}^N P(\hat{y}_i > t \mid y_i = 0) \times PV(ROI) - (\overline{ROI} + \bar{y} \times \overline{EAD} \times \overline{LGD}),$$

where $P(\hat{y}_i > t \mid y_i = 0)$ are the false positive cases, $\overline{ROI}$ is the average interest revenue for the portfolio, $\bar{y}$ is the average default rate for the portfolio, $y = (y_1, \dots, y_N)$, $\overline{EAD}$ is the average EAD and $\overline{LGD}$ is the average LGD.

Note that our proposed approach allows us to split false estimates above and below the decision threshold which provides additional insights on model performance. In addition, it considers both in terms of making judgements on model selection as well as the overall view. The proposed Financial Performance Metric (FPM) for calculates the misclassification cost by:

$$FPM = FNC + FPC. \quad (2)$$

4 Methods used

We utilise six widely used machine learning algorithms of varying complexity on a publicly available dataset (GCD) to evaluate which performs best according to our proposed FPM. The methods compared include: Logistic Regression, Random Forest, Gradient Boosting Machine, Decision Trees, Support Vector Machine and Artificial Neural Network Multilayer Perceptron. Next, we give a brief description of each algorithm and refer to the references provided for more details.

4.1 Logistic regression

Logistic Regression (LR) is the gold standard in credit risk modelling in industry and is typically the baseline model in credit scoring model comparison studies (Yhip & Alagheband, 2017). However, it has received criticism for its inability to detect non-linear or non-monotonic relationships. The LR is an extension of linear regression, using a logit function to make the distance between two binary points continuous. The logit function is a transformation between the linear model and the probability of the binary outcome. The LR is formulated as

$$\ln(p/1 - p) = \beta_0 + \beta_1 x_1 + \dots \beta_n x_n,$$

where p is the probability of the event occurring, β_0 is the y -intercept, and β_i , $i = 1, \dots, n$ are the coefficient of the independent variable x_i .

4.2 Decision tree

Decision Trees (DTRs) utilise a flowchart or tree-like decision making approach that is easily visualised from left to right and is composed of “burst” nodes split into different paths. The three types of nodes include Root nodes, which compile the entire sample and are then divided

into multiple sets; Decision nodes, which are typically represented by squares, represent sub-nodes that diverge further into further possibilities; and Terminal nodes, which represent the outcome that cannot be categorized further. The tree structure is built from the top down by selecting the best decision node to split first, and then after, based on Measures of entropy and information gain are used to select the best decision to node first and then subsequent decisions. Weights are calculated for each chance node by estimating the conditional (joint) probabilities. DTRs and variants thereof have been shown to be highly effective for credit risk modelling (Dumitrescu et al., 2022; Tian et al., 2020). DTRs are formulated as

$$(X, y) = (x_1, x_2, x_3, \dots, x_n, Y),$$

where $x_1, x_2, x_3, \dots, x_n$ are the independent variables, and Y is the target variable.

4.3 Random forests

Random Forest (RF) (Breiman, 2001) is a more complex bagging implementation of DTRs. It grows and combines multiple DTRs to create a “forest”. The RF has demonstrated superior model performance when applied to credit modelling research (Lessman et al., 2015). Formally, an RF is a predictor consisting of a collection of randomised base regression trees, as given below:

$$\bar{r}_n(X, D_n) = E_\theta[r_n(X, \theta_m, D_n)],$$

where E_θ represents the expectation taken with respect to the random parameter θ , conditioned on X and the dataset D_n . The parameters $(\theta_1, \theta_2, \dots, \theta_m)$ are independent and identically distributed (i.i.d.) realizations of the randomizing variable θ . Each tree is built independently by randomizing over subsets of the data or features, and the final output $\bar{r}_n(X, D_n)$ aggregates predictions across the ensemble.

4.4 Gradient boosting machine

Gradient Boosting Machine (GBM) is an implementation of boosting, an iterative technique which adjusts the weight of an observation based on the last classification (Friedman, 2001). GBM and its variants such as Extreme Gradient Boosting have also performed strongly in the comparative classification literature (Odegua, 2020). The main idea of GBM is to add new models to the ensemble sequentially. At each particular iteration, a new weak, base-learner model is trained with respect to the error of the whole ensemble so far. One can arbitrarily specify both the loss function and the base-learner models on demand. The general update rule in GBM can be expressed as

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x), \quad m = 1, 2, \dots, M,$$

where where $F_m(x)$ is the model prediction after m iterations (or trees), and

$$F_0(x) = \arg \min_{\gamma} \sum_{i=1}^n L(y_i, \gamma),$$

and decision tree $h_m(x)$ is fitted to the residuals by

$$h_m(x) \approx r_i^{(m)}.$$

Here, the pseudo-residuals is computed using

$$r_i^{(m)} = - \left[\frac{\partial L(y_i, \hat{y}_i^{(m-1)})}{\partial \hat{y}_i^{(m-1)}} \right],$$

where y_i is the actual class label of the i -th data point, $\hat{y}_i^{(m)}$ is the predicted probability of the i -th data point belonging to the positive class after m iterations, x_i is the i -th input feature vector, γ_m is the weight or learning rate for the m -th tree and controls the step size at each iteration, and L is the loss function.

4.5 Support vector machine

Support Vector Machines (SVMs) are supervised learning models which work by constructing hyperplanes in a multidimensional space to separates cases of different class labels (Vapnik, 1996). The samples that lie on boundaries of different classes are referred to as support vectors. The underlying principle behind SVM-based classification is to maximize the margin between the support vectors using kernel functions. SVMs have achieved superior performance in both retail (Obare & Muraya, 2018) and non-retail credit risk assessment (Teles et al., 2021). The SVM is formulated as

$$f(x) = \sum_{i=1}^P a_i K(x, x_i),$$

where P is the number of support vectors, a_i is an element of the parameter vector, x_i is a vector of regressors, K is a function referred to as the kernel and a_i is the number of parameters. The kernel can be a Radial Basis Function, polynomial or a two-layer neural network.

4.6 Multilayer perceptron

Multilayer Perceptron (MLP) is an Artificial Neural Network characterised by a back-propagation algorithm which uses a special class of feedforward networks (Baum, 1988). MLP has more than one hidden layer and the network moves only in the forward direction, with no loopback. These neural networks are good for both classification and prediction. MLP has been successfully deployed in credit risk modelling research achieving superior performance. MLP composes the final output function of a network using

$$f_k(x) = f_k(a_k(x)).$$

The function is applied to an input activation function consisting of $a_k(x)$. The input frequently comprises a computation of the form

$$a(x) = Wh(x) + b,$$

where x is the input vector, b is the bias vector, W is the weight matrix and h is the activation function output. The function $a(x)$ takes a vector argument and returns a vector as a result, note that $a_k(x)$ is just one of the elements of $a(x)$.

5 Datasets and numerical results

In this section, we describe GCD and Home Credit Dataset (HCD), the sampling procedure, the data modelling and the simulation process. This is followed by presenting the results of our numerical experiments.

5.1 Dataset and sampling

The publicly available German Credit Dataset (GCD) (Hofmann, 1994) is used in our experiments. It contains 1000 observations across 20 predictor variables and a response variable, indicating goods (i.e. negatives) and bads (i.e. positives). Due to confidentiality which is characteristic of financial data most research is carried out using publicly available datasets such as the GCD, which is among the most widely used public datasets for credit risk research (see for examples, Beque et al., 2017; Due & Graff, 2019; Khashman, 2010; Dong et al., 2010).

According to Ramezan et al. (2021), the size of the training data set is a major determinant in the classification performance. The authors compared the error of prediction across different sample sizes for a variety of algorithms including RF, GBM and MLP. They found that all these algorithms achieved more than 95% accuracy with a training sample of 625.

Alam et al. (2010) developed an approach for power estimation in LR to determine whether a parameter is significantly different to zero, combining the prior work of Whittemore (1981) and Hsieh et al. (1998). Their results show that a sample size of 655 is needed to achieve a statistical power of 0.9630.

These studies suggest that using a training sample of at least 655 will achieve strong accuracy for machine learning models and adequate statistical power for models incorporating hypothesis testing as per LR. Thus, the training sample of 712 used should be large enough to minimise the error of prediction and to determine the real significance of one or more parameters.

5.2 Modelling data

In this section, we describe modelling the GCD. Predictors in this data include: • Status of existing checking account • Duration in month • Credit history • Purpose • Credit amount • Savings account • Present employment since • Installment rate in percentage of disposable income • Personal status and sex • Other debtors • Present residence since • Property • Age in years • Other installment plans • Housing • Number of existing credits at this bank • Job • Number of dependents • Telephone • Foreign worker.

Prior to modelling, all predictor variables are transformed according to the Weights of Evidence (WoE) discretization approach which is widely used in industry. While LR is the only approach utilised that requires the monotonicity provided by WoE transformation, discretization was performed to ensure that data aligned to industry practice as much as possible. Note that making explanatory variables either strictly increasing or strictly decreasing (i.e. monotonic) is a critical requirement for scorecard models; as indicators for riskiness each retained variable must be linearly related to default for face validity in credit provision environments. The formula for WoE discretization is given below:

$$WoE_{x_i} = \ln \left(\frac{\% \text{ of } y = 0, \text{ where } x = i}{\% \text{ of } y = 1, \text{ where } x = i} \right).$$

Table 2 Gamma parameters for simulated EAD

Asset class	Shape	Scale
CC	3	5000
SL	3	50,000
Mo	3	400,000
LL/SME	3	1,000,000

5.3 Simulation of data

Next, we discuss the simulation of the GCD. Due to the lack of information necessary to evaluate cost parameters, simulated variables representing four different asset classes were incorporated into the GCD. Importantly, only the 20 original predictors and the target default variable were accessible to the model. This ensured that all models operated with identical default class information within each condition, whether training or testing. The simulated variables were used exclusively for cost calculations, which informed model selection decisions. To build on the GCD, we simulated the key missing variables required for cost analysis across the 1000 observations that comprise the underlying dataset.

Furthermore, we simulate EAD values according to a Gamma distribution (see, for examples Jimenez & Mencia, 2007; Assadsolimani & Chetalova, 2017) given below:

$$f(z) = \frac{z^{a-1} e^{-bz} b^a}{\Gamma(a)}, \quad z > 0, \quad a, b > 0,$$

where $\Gamma(a)$ is the gamma function, a is the shape parameter, and b denotes the rate parameter.

We use the Gamma distribution to simulate four different EAD ranges representing typical magnitudes of exposure observed in a selection of different economies. Our review of lending statistics from jurisdictions such as Australia (Australian Bureau of Statistics, 2024), the UK (Bank of England, 2024), and the USA (Ostrowski, 2024) revealed significant disparities both between and within these regions. For instance, in the USA, average mortgage lending exposures in California are over three times higher than those in Indiana. Further complicating generalizations across jurisdictions are factors such as variations in exposure pricing, exchange rates, regulatory practices, and the risk appetites of individual lenders. To ensure representative asset classes, we selected scale-appropriate ranges that encompass typical portfolios observed in Australia, the UK, and the USA.

The primary objective in evaluating these portfolios was to examine the influence of model metrics, including FPM, between portfolios of varying sizes. Specifically, we sought to determine whether larger interest payments influenced model selection decisions. The simulated ranges across different portfolio scales provided meaningful differentiation to support this analysis.

By modelling different exposures according to a Credit Card (CC), Small Loan (SL), Mortgage (Mo) and Large Loan/Small to Medium Enterprise Loan (LL/SME), we seek to observe the impact of different default and opportunity cost ranges. This follows from Lessman et al. (2015) who found that the impact of failing to consider the bias towards true positives was magnified the larger the cost of a false positive.

After setting the seed to ensure values are reproducible, EAD distributions were simulated using the parameters given in Table 2.

Table 3 Descriptive statistics of simulated assets (GCD)

Asset class	Mean	Median	Standard deviation
CC	\$12,039	\$10,419	\$7134
SL	\$152,263	\$139,396	\$86,212
Mo	\$1,185,477	\$1,034,473	\$698,810
LL/SME	\$2,996,111	\$2,679,932	\$1,723,469

Table 4 Observed counts and default rate (GCD)

Dataset	Good	Bad	Total	Default (%)
Training	502	210	712	29.49
Test	198	90	288	31.25

The proportion of the loan already paid (used to calculate Loan principal) is simulated according to a Beta distribution, that is:

$$f(z) = \frac{1}{\beta(a, b)} z^{a-1} (1-z)^{b-1}, \quad 0 \leq z \leq 1,$$

where $\beta(a, b)$ is the beta function, $a > 0$ and $b > 0$ are shape parameters. The Beta distribution was simulated with both shape parameters equal to one. In the asymptotic sample this translates to a roughly uniform distribution between zero and one with some random variation.

Monthly payment (Pmt) amounts are calculated according to the payment equation detailed in Finlay (2009) as

$$Pmt = \frac{P_r \times r/12 \times (1 + r/12)^{12t}}{(1 + r/12)^{12t} - 1},$$

where P_r = Principal, r = Annual interest and t = Term of the loan in years.

We also keep the underlying data as realistic as possible according to the aforementioned hypothetical asset classes, basing distributions on real portfolios observed at Australian and European banks. Despite these efforts, some differences are still apparent in the data. The relatively high default rate observed in the GCD clearly sets it apart from datasets observed in industry. However, given well-known constraints with classification modelling it is common practice to perform sampling on credit risk data such that the minority class (ie defaults) is markedly increased. Therefore, we can simply view the larger than normal minority class as being the result of prior sampling to address the class imbalance. In Table 3, we provide descriptive statistics for the four simulated asset classes; CC, SL, Mo and LL/SME.

In Table 4, we provide the split between the training and test datasets. It includes counts of good (non-defaulting) and bad (defaulting) observations as well as the percent of each outcome condition. Random sampling between the training and test datasets led to slight variations in defaulting proportions. Initial testing revealed no significant differences in results between equally sampled classes and randomly sampled ones. Therefore, to maintain realistic conditions, we opted for the randomly sampled approach, which is commonly used in industry practices.

A final potential deviation from reality relates to the credit card asset class, whereby it was treated like a one-off loan held for a set number of months. This is not realistic since on average credit card customers tend to draw down credit continuously, implying that we may

Table 5 Descriptive statistics of Home Credit Dataset (HCD)

Asset class	Mean	Median	Standard deviation
Various product types	\$1,203,458	\$1,069,602	\$697,163

Table 6 Observed counts and default rate (HCD)

Dataset	Good	Bad	Total	Default (%)
Training	197,910	17,448	215,258	8.10
Test	84,876	7377	92,253	7.99

have under represented the interest which might normally be collected over the life of this asset class.

To further demonstrate the robustness of the proposed FPM, we applied it to the HCD also used in other studies, such as Wang et al. (2020). This data, available on Kaggle (Montoya et al., 2018), highlights Home Credit strives to expand financial inclusion for the unbanked population, featuring a distinct loss structure unlike typical lending portfolios. Since the HCD encompasses various lending products with different loss dynamics, this introduces some noise into predictions. However, it contains information similar to the GCD, covering a larger dataset of 307,501 rows in the fully labelled subset.

For the comparison purpose, we analysed a single portfolio consisting of various product types in the HCD rather than multiple single asset classes, using the same simulation approach as with the GCD. In Table 5, we provide descriptive statistics for the non-specific portfolio from the HCD.

Table 6 displays information on the counts of good (non-defaulting) and bad (defaulting) observations, the percentage of each outcome condition and the rate of default in the training and test splits of the HCD.

5.4 Implementation

The final decision prior to modelling was to either optimise thresholds according to specific models which would have enhanced realism yet reduced comparability, or to hold the threshold constant at $t = 0.5$ to focus on the differential capabilities of each algorithm and the performance metrics; we opted for the latter. Furthermore, while preliminary model tuning produced some small increases in model statistical performance, it came at the cost of over-fitting to the training dataset. Thus, as all six models achieved acceptably strong performance according to the statistical metrics only minor model tuning was performed. Statistical metrics were calculated first using automated outputs from R software and then recalculated manually according to the relevant equations.

5.5 Results

In this section, we present the results of our numerical experiments. We start by comparing the performance of our proposed cost metric with that of the commonly used approach by Bahnsen et al., and noting differences and similarities in model ranking between the two approaches.

Table 7 False prediction costs using Bahnsen's approach (CSEDCM) as a percentage of proposed cost metric (FPM), Eq. (3)

Threshold (cost)	Methods	CC (%)	SL (%)	Mo (%)	LL/SME (%)
Default	MLP	1.04	7.67	7.10	8.09
Default	DTR	1.29	8.20	8.50	11.34
Default	GBM	1.02	7.05	7.63	9.14
Default	LR	1.21	9.77	7.20	10.76
Default	RF	1.20	8.98	8.21	13.87
Default	SVM	1.28	5.74	11.22	15.51
Opportunity	MLP	30.72	43.43	38.09	40.50
Opportunity	DTR	41.15	43.50	33.34	35.01
Opportunity	GBM	25.56	45.47	31.16	38.96
Opportunity	LR	30.91	49.62	44.61	46.39
Opportunity	RF	31.74	44.61	39.45	22.33
Opportunity	SVM	29.88	14.06	29.94	45.09
Overall	MLP	17.30	33.63	29.61	30.87
Overall	DTR	19.26	27.54	21.59	24.11
Overall	GBM	13.48	33.08	22.40	27.65
Overall	LR	13.62	35.54	30.01	32.63
Overall	RF	11.70	26.39	22.56	17.21
Overall	SVM	21.51	10.95	24.61	39.91

We calculated the cost of false predictions using both the proposed FPM and that put forward by the CSEDCM. Noting that all of CSEDCM's cost estimates were smaller than the FPM, we then calculated the percentage for each model and threshold position using the following equation:

$$Cost\% = \frac{CSEDCM}{FPM} \times 100. \quad (3)$$

In Table 7, we report the results obtained using this equation. The first two columns in this table are the cost types and the six algorithms used in our comparison, and the last four columns are the simulated asset classes. The results show that Bahnsen's approach routinely underestimates the true costs of false predictions, with under prediction sometimes as different as one hundredth of the proposed FPM. For example, in estimating the default cost of false prediction for the GBM, Bahnsen's estimate was just 1.02% of the FPM.

The huge disparity in false prediction costs drastically increases the likelihood for inadvertently selecting the wrong model under the belief it presents the lowest cost of misclassification. We've further enhanced the model selection process by demonstrating performance measurement differentials on either sides of the decision boundary, whereas most industry practitioners as well as researchers consider only the overall view, and that is usually done comparing averaged statistical metrics such as Gini and AUC. Based on the cost of false prediction estimated by the CSEDCM compared to the cost of false prediction estimated by the FPM from this table, we will utilise our proposed cost metric FPM for further analysis.

If the proposed FPM is utilized to estimate the cost of false prediction (the results presented in Table 8), the MLP demonstrates the lowest misclassification costs across all four asset classes (CC, \$17.4m; SL, \$42.2m; Mo, \$260.7m; LL/SME, \$488.7m), and is therefore the

best performing model. By contrast, if the CSEDCM was used for model selection, the RF would be selected for the CC (\$2.6m) and LL/SME (\$91.8m) asset classes and the SVM for small loans (\$5.1m) and DTR for mortgages (\$72.1m). Interestingly, when we peel back the cover to examine performance on either side of the decision boundary, both cost approaches favor the MLP for default cost, and RF performs best for opportunity cost, yet the considerable differences in the magnitude of false prediction costs result in differences at the aggregate level across the decision boundary, which is the level at which the CSEDCM approach operates at.

Now, we discuss the comparison between each respective method according to the financial cost of losses calculated on false predictions. The losses associated with false negatives (applicants the model predicts as good but who later default; the default cost), as well as false positives (applicants the model predicts as bad yet who do not default; the opportunity cost and mostly comprise the interest revenue foregone by rejecting an otherwise good applicant), are included. Separating observations into false positives versus false negatives required splitting the sample above and below the threshold which enhanced comparisons with the statistical metrics.

The comparison results, the costs associated with individual components and the overall FPM, are provided in Table 8. The 'Above' column represents false positive misclassification costs, incurred when the model incorrectly classifies individuals as defaulters who did not default. The 'Below' column captures false negative misclassification costs, summing the losses from applicants classified as low risk but who ultimately defaulted. Finally, the 'Overall' column represents the total misclassification costs, combining both false positives and false negatives, comprising the proposed metric.

Next, we provide the results of the classical statistical performance metrics in Table 9, where the first two columns show the six methods used and the threshold split, and the last three columns are the statistical metrics. Typically these are only presented at the overall level, which is effectively an average across all observations, however we additionally estimate performance measures above and below the threshold. Splitting the performance metrics in this manner provides further insight into each method and how they perform on either side of the decision threshold as well as the measurements themselves.

The results from Table 9 show that when we rank each algorithm by statistical performance seeking to identify the best method, we see that the method rankings are very different comparing either side of the threshold with overall. For example, Gini shows that the GBM performed best above the threshold while the LR was superior below, yet it was the MLP that came out on top according to this metric.

Further, the Gini and AUC result in identical rankings, reinforcing the notion that the practise of presenting both metrics together is redundant. By contrast, the accuracy of methods almost always ranks methods differently to Gini/AUC, reflecting the different philosophy in taking the proportion of true predictions from all predictions, versus discriminating between goods and bads using the number of concordant pairs.

The routine use of these statistical metrics in isolation is problematic because as has already been established, while they may be useful in distinguishing between a good and a bad model, they are not useful for making selections between good models. Moreover, they do not provide insight on the real parameter of interest which is minimising the false predictions, in particular their cost in a credit scoring environment. Furthermore, 'overall' estimates effectively average across the decision boundary thus obscuring important details of model performance.

We note that when used in the traditional manner assessing statistical performance overall, the MLP was the clear winner overall, yet only marginally stronger than the LR. The marginal

Table 8 Financial cost metrics (per \$1000) by methods, asset class and threshold using test dataset (GCD)

Method	Assets	Above	Below	Overall
DTR	CC	\$11,233	\$13,686	\$24,919
GBM	CC	\$10,656	\$10,335	\$20,991
LR	CC	\$8134	\$11,337	\$19,471
MLP	CC	\$9541	\$7882	\$17,423
RF	CC	\$7749	\$14,784	\$22,533
SVM	CC	\$13,662	\$5652	\$19,314
DTR	SL	\$30,664	\$25,311	\$55,975
GBM	SL	\$38,119	\$18,163	\$56,283
LR	SL	\$29,245	\$15,973	\$45,218
MLP	SL	\$30,607	\$11,557	\$42,164
RF	SL	\$25,031	\$26,199	\$51,230
SVM	SL	\$29,145	\$17,368	\$46,513
DTR	Mo	\$176,080	\$158,013	\$334,093
GBM	Mo	\$204,541	\$121,299	\$325,840
LR	Mo	\$180,241	\$115,443	\$295,684
MLP	Mo	\$189,427	\$71,316	\$260,743
RF	Mo	\$155,153	\$182,587	\$337,741
SVM	Mo	\$242,131	\$96,377	\$338,508
DTR	LL/SME	\$314,838	\$268,673	\$583,511
GBM	LL/SME	\$402,095	\$245,552	\$647,647
LR	LL/SME	\$324,782	\$204,305	\$529,087
MLP	LL/SME	\$343,546	\$145,248	\$488,795
RF	LL/SME	\$210,919	\$322,532	\$533,451
SVM	LL/SME	\$537,818	\$114,113	\$651,931

statistical improvement obtained from the MLP would usually be discarded in favour of the simpler and more easily implemented and understood LR. Indeed, when used in conjunction with the FPM, the true cost of selection decisions can be easily understood.

In addition, the differential performance on either side of the threshold provides some insight into how each algorithm works, and, where each performs strongest. For instance, it is notable that according to ACC the SVM performed amongst the best below the threshold, yet it was also the worst above the threshold. The observed variation in method ranking between model metrics underscores the notion that using traditional performance measures blindly results in suboptimal method selections. Certainly averaging metrics across the decision boundary obscures important details on performance. Thus, the traditional performance measures present challenges for method selection and must be used with caution.

From the results, a notable observation supports the findings by Verbraken et al. (2015), whereby the larger the magnitude of the principal loan amount, the more important that opportunity costs become. Given the scale of the largest asset class we simulated and the consequent magnitude of interest revenue that would be lost, our findings show that for these data opportunity costs are larger than default costs. The disparity between opportunity and default cost is largest where the initial exposure is smallest, in the credit card asset class, and diminishes as the average size of the asset class increases. This suggests that the larger the

Table 9 Statistical metrics by threshold and methods using test dataset (GCD)

Method	Threshold	Gini	ACC	AUC
MLP	Above	0.3273	0.7835	0.6636
DTR	Above	0.3906	0.7619	0.6953
GBM	Above	0.4131	0.7553	0.7065
LR	Above	0.3681	0.8046	0.684
RF	Above	0.3279	0.7922	0.6639
SVM	Above	0.3628	0.71	0.6814
MLP	Below	0.8358	0.9319	0.9179
DTR	Below	0.8224	0.8725	0.9112
GBM	Below	0.8224	0.8969	0.9112
LR	Below	0.8558	0.9005	0.9279
RF	Below	0.8014	0.8863	0.9007
SVM	Below	0.7682	0.9198	0.8841
MLP	Overall	0.8914	0.8819	0.9457
DTR	Overall	0.8517	0.8403	0.9259
GBM	Overall	0.8561	0.8507	0.9281
LR	Overall	0.8903	0.8715	0.9452
RF	Overall	0.8523	0.8611	0.9262
SVM	Overall	0.8392	0.8467	0.9196

credit contract, the more careful lenders must be to avoid misclassifying a good credit risk as bad.

However, this is in a limited sample of 1000 customers, whereas in a real credit portfolio products like CC and SL customers may number in their millions. While our findings support the notion that opportunity costs become increasingly important the larger the principle. However, given the volume of low value lending products like CC this may not hold, particularly in unsecured lending in which the LGD tends to be significantly higher.

From a method selection perspective, the algorithm resulting in the smallest loss was the winner, in line with the profit-based motive which characterises commercial enterprise. Results show that the RF posed the lowest misclassification cost across all asset classes related to false positives, that is for applicants that the model has scored above the decision threshold who do not default. By contrast, the SVM posed the lowest misclassification costs for two asset classes (CC and LL). The MLP misclassification cost was lowest for SL and Mo for applicants who the method approves, that is applicants who score below the decision threshold, yet who later default. Overall, the MLP performed the best across all four asset classes.

We conclude by discussing the power curve that is used to empirically determine the statistical power in our training sample. More precisely, the curve is produced to determine whether the GCD provides adequate statistical power with which to make inferences about model performance. While hypothesis testing is only a feature of statistical modelling, and therefore, only relates to the LR algorithm, it is instructive for comparison purposes with the other methods investigated. Note that the statistical power was calculated using two assumed target variable probabilities, the required inputs for this calculation. However, under both probability assumptions, the statistical power approached 100% prior to the training sample reaching 700. Thus, the power curve demonstrates that the training sample selected at random

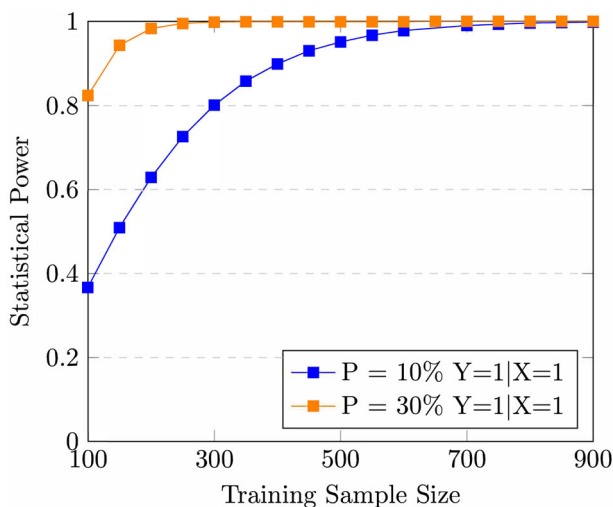


Fig. 1 Sample size power curve for classification

from the GCD was sufficiently large to provide adequate statistical power with which to estimate one or more parameters using LR. Figure 1 depicts the results of the power analysis.

Furthermore, as machine learning algorithms do not formally perform hypothesis testing as statistical models do, true power analysis is not possible. Instead, we calculate the error of prediction over a range of training sample sizes to test whether the sample size used was large enough. The determination of the adequacy of the size was made by observing whether the error rate was still declining as the training sample grew, or whether it flattened out. Each model was tested on each of nine training samples extracted from the GCD ranging in size from 100 to 900. After each subset was extracted to form the training data, the remaining proportion comprised the test data. Thus, where the training dataset was 100, the test dataset was 900; where the training dataset was 200, the test dataset was 800, and so on until the maximum split whereby the training dataset was 900 and the test dataset was 100. The sample size versus error rate charts for all six methods are shown in Figs. 2, 3, 4, 5, 6 and 7, respectively. Note that in Fig. 1, the power greater than 0.8 is generally considered acceptable, while for Figs. 2, 3, 4, 5, 6 and 7 the lower the error rate the better quality the method.

In early method testing, model tuning appeared to improve the fit to the training dataset yet reduced performance on the test data. Thus, to minimise overfitting to the training dataset only basic model tuning was performed. The sample size by error plots for each respective model reveals that the training and test datasets achieved similar performance irrespective of sample size, with only the smallest training sample size of 100 (test = 900) resulting in a maximum error of prediction approaching 0.2.

The two methods which showed elevated error rates were the GBM and the SVM. Contrary to the converging performance between training and test datasets observed for the other four methods, the error of prediction for these methods were substantially larger for the test dataset irrespective of its relative size in comparison to the training data. For example, for the GBM there is at least a 10% difference in performance even when the training data was at its maximum of 900.

While performance never converges across any sample size it is noteworthy that the error rate in the test data is generally declining for both the GBM and the SVM as the size of the

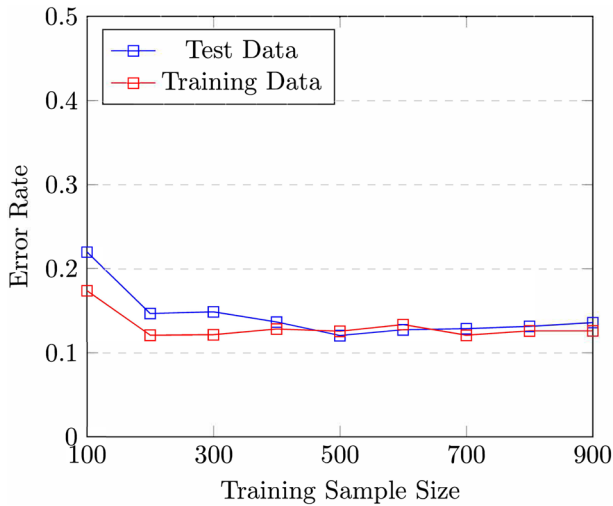


Fig. 2 LR sample size versus error rate

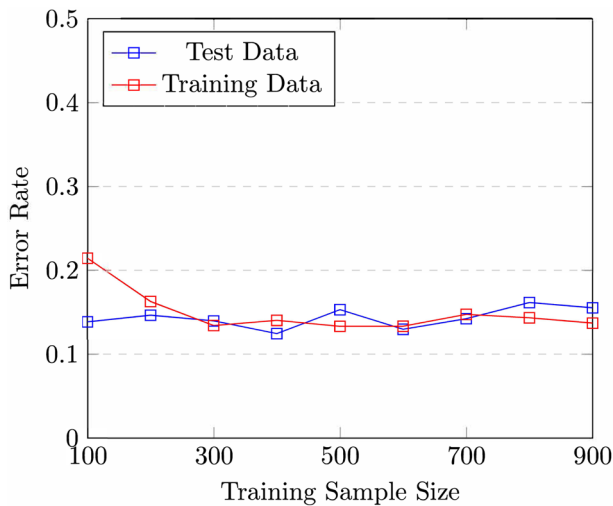


Fig. 3 RF sample size versus error rate

training dataset increases, and relatively flat for the other four methods. While performance may have continued to improve if the training sample was even larger, it implies that the GBM and the SVM may require larger training datasets to control the error in test data. An alternative interpretation is that the GBM and the SVM formulations did not generalise to this specific dataset as well as the other algorithms.

Similar levels of the performance for both test and training datasets were observed for LR, RF, MLP and DTR, yet for the GBM the actual magnitude of difference in error was only around 0.1 by the time training data was around 700, and roughly 0.06 for the SVM. Taken together these results support the notion that the training dataset was sufficiently large to produce reliable results for all algorithms tested.

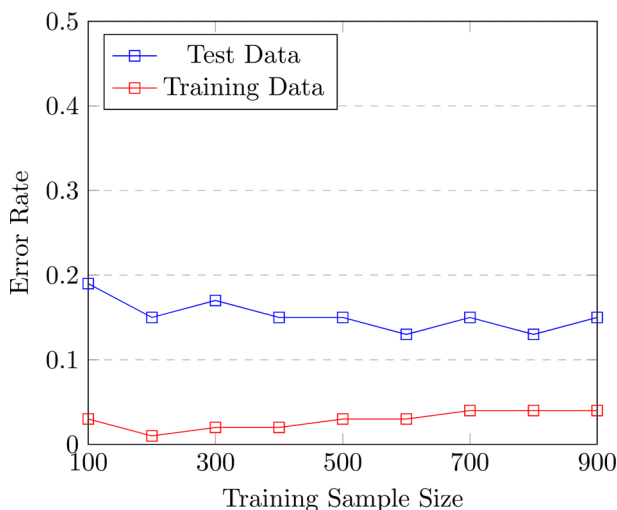


Fig. 4 GBM sample size versus error rate

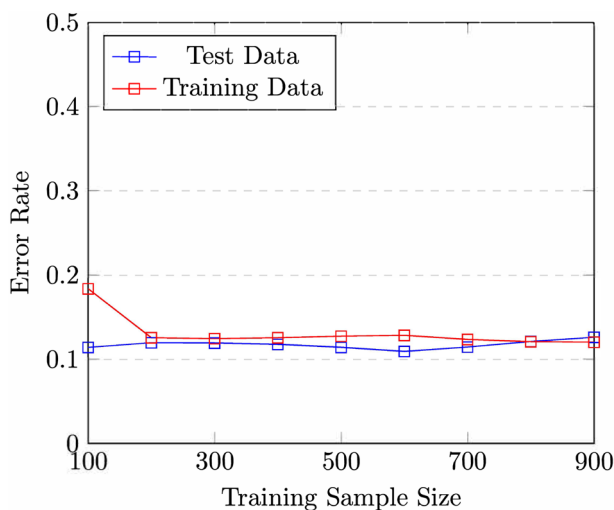


Fig. 5 MLP sample size versus error rate

As mentioned earlier, we used the HCD with a single portfolio of various product types to further demonstrate the robustness of the FPM.

Table 10 presents model performance on the HCD including a statistical accuracy metric and the difference from the LR algorithm for reference. Using a classical statistical performance approach, SVM would typically be chosen. However, the accuracy differences among the six algorithms are trivial. In contrast, the financial losses from false predictions in the FPM reveals significant differences across algorithms. This is consistent with the observations for the GCD. The FPM identified RF as the best-performing model, closely followed by the MLP. The results also indicate that utilizing the FPM would achieve a significant saving of approximately \$321 million and \$297 million over the LR, respectively.

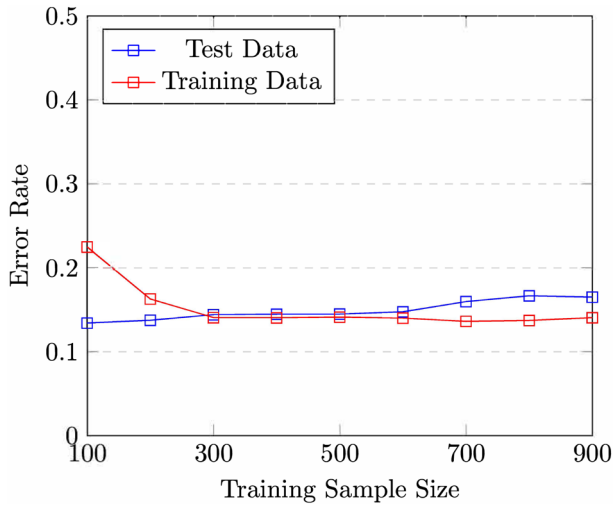


Fig. 6 DTR sample size versus error rate

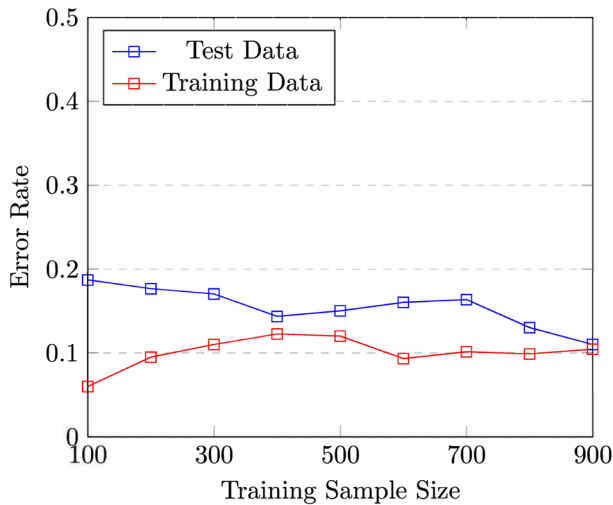


Fig. 7 SVM sample size versus error rate

Table 10 Financial cost metrics (per \$1000) by methods using test dataset HCD

Method	Accuracy	Above	Below	Overall	Differ from LR
DTR	0.9197	\$353,830	\$635,740	\$989,570	−\$198,240
GBM	0.9201	\$310,680	\$498,620	\$809,300	−\$17,970
LR	0.9200	\$318,150	\$473,180	\$791,330	–
MLP	0.9200	\$204,900	\$289,330	\$494,230	\$297,100
RF	0.9200	\$138,990	\$330,380	\$469,370	\$321,960
SVM	0.9155	\$185,810	\$316,250	\$502,060	\$289,270

6 Discussion and conclusion

In this section, we provide discussion on the results presented in this paper and conclude with future research directions.

Publicly available datasets enable industry practitioners to emulate the results achieved in published literature more easily, thus expediting the uptake of new research. This study shows that even relatively small, publicly available datasets like the GCD and the HCD are not only large enough to provide sufficient statistical power with which to train statistical models but also to minimise the error of prediction arising from the application of machine learning algorithms on training samples anywhere above 100 observations.

Moreover, we demonstrated that where necessary financial information is not available, it can be simulated to provide insights on cost dynamics as a consideration in model selection particularly where these are calibrated to real lending portfolios. While our cost dynamics are estimated on simulated data, the results from tests on statistical power and error of prediction show that these samples are sufficiently large to avoid sampling bias and therefore, make inferences on financial losses.

Furthermore, by splitting results on either side of the decision threshold, we showed how model performance differs according to which algorithm has been used, thus providing additional insights into algorithm selection and performance metrics alike. The model ranking according to the classical performance metrics varied greatly when split in this manner.

The present study has focused on individual algorithms. However, as model performance and ranking were highly variable on either side of the decision boundary, a stacking approach may well outperform single algorithms. By combining the strengths of algorithms on either side of the threshold it may be possible to eliminate or at least mitigate their weaknesses to develop a stronger overall prediction system.

In addition, building on previous research, we have sought to propose the most realistic estimates of false prediction costs yet. Although we utilised relatively small datasets of few observations, the proposed FPM could be extended and applied to larger lending portfolios which could potentially provide a significant financial benefit to organizations.

Taking the example of the mortgage asset class, the best performing algorithm overall was, for example, the MLP in the GCD, and selecting that over the next best algorithm (LR) resulted in a benefit of approximately 35 million on a dataset of <300 applicants. While this is a large enough difference to be meaningful in selecting the most fit-for-purpose model, a real-life mortgage portfolio could easily be a thousand times larger. Applying the ratio of 1:1000 puts the value of selecting MLP over LR at \$35 Billion, a massive financial benefit for even the largest corporations trading on a global scale.

Furthermore, we have noted the extant issues with the three most popular performance metrics, AUC, Gini, and ACC. Prior research suggests these measures are unsuitable for model selection. For example, although AUC may be able to differentiate a good model from a bad one it is unable to distinguish between good ones. While it is coherent at separating class membership, it is also incoherent on the costs of those classes, and the actual loss resulting from false predictions. Although these incoherent approaches may be useful as a hurdle to differentiate between good and bad candidate models, the financial loss should be used for identifying the best performer. Therefore, we propose an optimal system whereby once candidate models achieve a minimum performance threshold, the financial loss measures should be used to select the best-performing model.

Using the proposed Financial Performance Metric (FPM) enables industry practitioners to select models which best align with their motives for using the models in the first place for

profit. As it is demonstrated herein for credit scorecards, the same principles can be applied to any commercial application of classification models in which the loss function is more nuanced than simply separating goods and bads.

Most of the published credit modelling research focuses on application scorecards, nevertheless, they are not the most material models used by banks globally. Future research should deeply examine behavioral models which measure the ongoing risk once applicants become customers. Both point in time, as well as through the economic cycle models are important considerations required by prudential authorities. These models have a huge impact on not only retained solvency capital, but also the overall health of the financial system. We tested our findings on a comparison dataset, but further testing including replacing simulated variables with real-time measurements may serve to underscore the value of using the financial performance metric. Lastly, there is a dearth of published research on optimal monitoring solutions for credit models over time, which is an urgent priority for industry.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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